Systemic Risk in Multiplex Financial Networks: Centrality, Contagion, and Policy Intervention

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Abstract

The 2008 global financial crisis highlighted the limitations of analyzing systemic risk through aggregated single-layer networks. Traditional models often obscure the interrelated exposures among financial institutions between different instrument types. This thesis investigates the propagation of systemic risk by modeling the global banking system as a multi-layer network, using historical data from the Bank for International Settlements (BIS) during the 2008 crisis.

To identify systemically important nodes, we apply MultiRank, a centrality metric designed for multi-layer topologies. We subsequently introduce a dual-channel contagion model that incorporates both solvency shocks and liquidity crunch. By assigning heterogeneous capital buffers based on topological centrality, we simulate how disparities in leverage between core hubs and peripheral nodes drive localized defaults into global cascades.

The simulation results demonstrate that the liquidity channel acts as the primary accelerator of systemic collapse. Furthermore, policy intervention analysis indicates that targeted liquidity injections into high-MultiRank hubs (e.g., the United States or Great Britain) yield a strictly higher preservation of global economic mass compared to untargeted interventions. Ultimately, this research provides a quantitative, graph-theoretic framework for evaluating systemic fragility and optimizing regulatory interventions in complex financial networks.

Contents

1 Introduction

1.1 Motivation

In modern financial markets, institutions are deeply interconnected through a myriad of capital transactions, including but not limited to interbank lending, derivatives trading, and cross-holding exposures. To systematically understand and analyze these multifaceted interactions, it is essential to employ a rigorous quantitative framework. By leveraging graph theory, we can abstract and characterize the complex topology of the entire financial system.

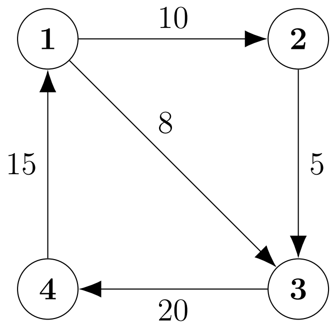
This mathematical representation naturally raises a series of critical research questions:

- How do we rigorously construct these multiplex financial networks?
- What are the underlying mechanisms governing the transmission and contagion of systemic risk within them?
- Ultimately, how can the development of such quantitative models inform and optimize policy interventions?

1.2 Definitions and Core Concepts

To quantitatively analyze systemic risk, we abstract the financial system into a weighted directed graph. The fundamental components of this network are formally defined as follows:

- **Nodes (Vertices V):** Each node represents an individual financial entity (e.g., Node i, Node j) operating within the global financial system.
- **Directed Edges (E):** The edges represent the credit exposure between nodes. A directed edge $i \to j$ exists if node i has a financial claim against node j. Crucially, while initial money flows from node i to node j (e.g. through interbank lending), the risk exposure points from node i to node j, because node i is now exposed to the potential default of node j.
- **Edge Weight (W):** For any existing directed edge $i \to j$, the weight W quantifies the exact dollar volume of the credit exposure.
- **Adjacency Matrix (A_{ij}):** The topology of the entire global banking system is mathematically compressed into an asymmetric adjacency matrix A. The elements of this matrix, denoted as A_{ij}, represent directed exposures, where $A_{ij} > 0$ indicates an active financial claim of node i against node j.



$$A = \begin{pmatrix} 0 & 10 & 8 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 20 \\ 15 & 0 & 0 & 0 \end{pmatrix}$$

Figure 1: An illustrative 4-node directed network and its corresponding adjacency matrix A. The element $A_{12} = 10$ indicates Node 1 has a credit exposure of 10 to Node 2.

2 Problem Formulation and Conceptual Framework

2.1 Limitations of Traditional Analysis

The global financial system is fundamentally a network of direct transactional exposures. Traditional frameworks, such as the Capital Asset Pricing Model (CAPM), evaluate systemic stability through the statistical correlation of asset returns. However, historical price correlation does not capture true structural dependency.

For example, two financial institutions may exhibit low equity correlation, suggesting statistical independence. If Institution A relies on Institution B for an overnight credit line, a default by B immediately paralyzes A. This direct counterparty linkage represents a severe structural risk. By evaluating stability solely through aggregated asset volumes and statistical metrics, traditional analyses inherently mask the underlying topology of vulnerability.

2.2 The Multiplex Paradigm

Connections among financial entities in the real world are never monolithic or homogeneous. Instead, they are constructed from multiple mutually coupled relationships of differing natures, which encompass interbank lending channels, joint asset holdings, and derivative contracts. This inherent *multiplexity* dictates that financial connections are not solitary threads but intertwined layers existing simultaneously.

Although each relationship connects nodes within the network, their economic implications differ significantly. Lending relationships facilitate credit sharing, joint positions can transmit fire-sale risks, and derivatives contracts can amplify counterparty risks. Compressing these distinct edges into a single-layer structure obscures the critical differentiation between these risk vectors. In reality, systemic crises rarely erupt in isolation. They arise from the coupling between layers: an initial shock to one layer propagates through behavioral adjustments or asset price shifts to another, ultimately triggering widespread chain reactions. The significance of the multiplex paradigm lies not only in decomposing networks into multiple layers but in emphasizing that financial systems are complex wholes composed of interdependent and mutually reinforcing strata.

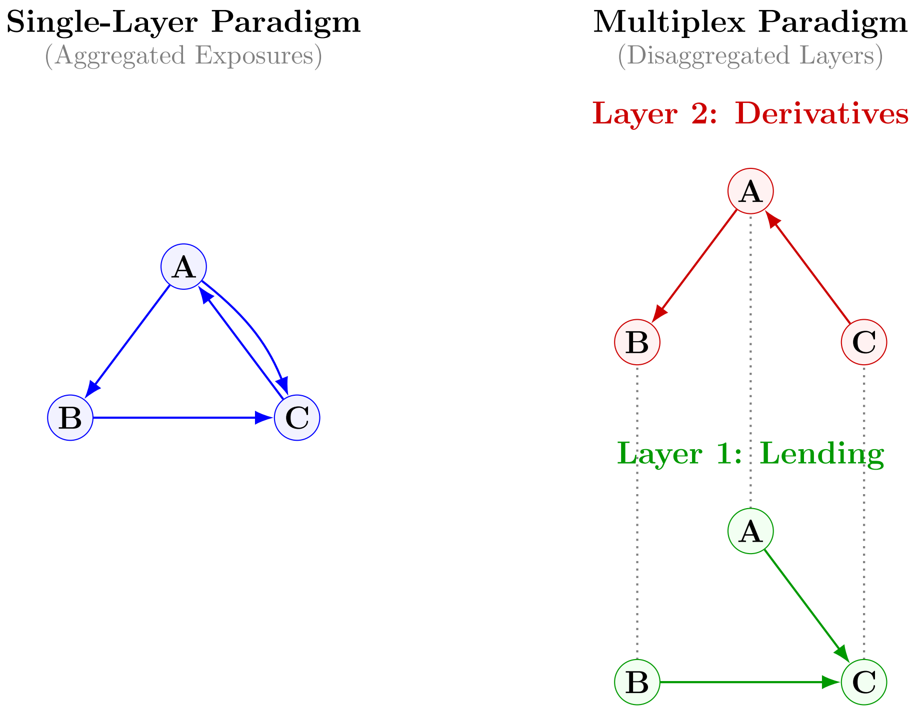


Figure 2: Conceptual comparison: A traditional aggregated single-layer network vs. a multiplex financial network. In the single-layer view (left), all connections are homogenized. In the multiplex view (right), identical nodes (e.g., Bank A) exist across multiple coupled layers, each representing a distinct type of financial exposure.

2.3 Core Research Objectives

Given the structural limitations of single-layer analysis and the necessity of the multiplex paradigm, this thesis identifies three core research objectives:

- **Metric (Centrality):** To identify true "core" nodes within a multi-layer network using the MultiRank algorithm.
- **Dynamics (Contagion):** model the propagation of financial shocks via a dual-channel contagion mechanism combining solvency shock and liquidity crunch.
- **Policy (Intervention):** Maximize systemic survival by targeting regulatory bailouts based on MultiRank centrality.

3 Mathematical Framework of Systemic Centrality

Before extending our analysis to multiplex networks, it is essential to establish the mathematical foundation of systemic importance within a single-layer topology. We adopt the PageRank algorithm—originally designed to rank network nodes based on random walks—as our baseline measure of node centrality. The formalization in this section follows the framework detailed by Bianconi (2018) in *Multilayer Networks: Structure and Functions*.

3.1 The Foundation: Single-Layer PageRank

A fundamental intuition behind PageRank is its built-in *buffering mechanism*. While it is reasonable to assume that being pointed to by a highly central node confers high centrality, this effect is diluted if the central node has a vast number of outgoing links. In a financial context, the risk or influence transmitted by a major institution is distributed across its entire portfolio of exposures.

Let $a = (a_{ij})$ be the $N \times N$ adjacency matrix of the financial network, where $a_{ij} > 0$ indicates a directed credit exposure from node i to node j for $i = 1, 2, \ldots, N$. To account for the buffering mechanism, the standard PageRank centrality x_i for a node i is defined as the steady-state solution to the following balance euqation:

$$x_i = \mu \sum_{j=1}^{N} \frac{a_{ji}}{k_j^{out}} x_j + w \tag{1}$$

where:

- $K_j^{out} = \sum_{i=1}^{N} a_{ji}$ is the standard out-degree (total outward exposure) of node j.
- $k_j^{out} = \max(K_j^{out}, 1)$ is the generalized out-degree, formulated to prevent division by zero for dangling nodes (nodes with no outward exposures).
- $\mu \in (0, 1)$ is the damping factor. In the random walker interpretation, a walker at a node with a non-zero out-degree hops to a random neighbor with probability μ, and teleports to any random node in the network with probability $1 - \mu$. A typical trade-off value utilized in standard topological analysis is $\mu = 0.85$.
- w is the teleportation term, ensuring the network remains strongly connected.

The teleportation term w is explicitly given by:

$$w = \frac{1}{N} \sum_{j=1}^{N} \left[(1 - \mu) + \mu \delta(K_j^{out}, 0) \right] x_j \tag{2}$$

where $\delta(x, y)$ indicates the Kronecker delta. This equation governs the boundary conditions of the random walk: if the walker reaches a node with zero out-degree ($\delta(K_j^{out}, 0) = 1$), it cannot traverse locally and instead jumps with a probability of one to a random node in the network. This ensures no probability mass (or financial shock impact) is permanently trapped in dead-end nodes.

Derivation of the Eigensystem

To understand the global dynamics and solve for the steady state, we transition from the scalar definition to a matrix representation. We define the random walk transition matrix P^{PR} as:

$$P_{ij}^{PR} = \frac{a_{ji}}{k_j^{out}} \tag{3}$$

By substituting Equation (2) and the transition matrix P^{PR} back into Equation (1), we express the centrality of node i entirely as a summation over all nodes j:

$$x_i = \sum_{j=1}^{N} \left(\mu P_{ij}^{PR} + \frac{1}{N} \left[(1-\mu) + \mu\delta(K_j^{out}, 0) \right] \right) x_j \tag{4}$$

To rigorously define x as an eigenvector, we absorb the teleportation term into a single matrix. Let us define the Matrix G with elements:

$$G_{ij} = \mu P_{ij}^{PR} + \frac{1}{N} \left[(1-\mu) + \mu\delta(K_j^{out}, 0) \right] \tag{5}$$

Here, the matrix G (often referred to in network science as the Google Matrix) is not just an algebraic substitution but carries a concrete probabilistic interpretation. The element G_{ij} represents the absolute total probability that a random walker transitions from node j to node i in a single time step. Couples two distinct network behaviors: the probability that the shock propagates along a direct edge (μP_{ij}^{PR}) and the probability of a systemic random jump ($\frac{1}{N} \left[(1-\mu) + \mu\delta(K_j^{out}, 0) \right]$).

To prove that G is a strictly left stochastic matrix, we must show that the sum of the elements in any column j is exactly 1. Let us compute the column sum $\sum_{i=1}^{N} G_{ij}$:

$$\begin{aligned} \sum_{i=1}^{N} G_{ij} &= \sum_{i=1}^{N} \left(\mu P_{ij}^{PR} + \frac{1}{N} \left[(1-\mu) + \mu\delta(K_j^{out}, 0) \right] \right) \\ &= \mu \sum_{i=1}^{N} P_{ij}^{PR} + \frac{1}{N} \left[(1-\mu) + \mu\delta(K_j^{out}, 0) \right] \sum_{i=1}^{N} 1 \end{aligned}$$

Knowing that $\sum_{i=1}^{N} 1 = N$, the equation simplifies to:

$$\sum_{i=1}^{N} G_{ij} = \mu \sum_{i=1}^{N} P_{ij}^{PR} + (1-\mu) + \mu\delta(K_j^{out}, 0)$$

To evaluate this expression, we consider the two distinct structural states for node j:

Case 1: Node j has outgoing edges ($K_j^{out} \geq 1$)
In this routine case, the indicator function $\delta(K_j^{out}, 0) = 0$. By the definition of the random walk transition matrix, the sum of probabilities transitioning out of node j is strictly 1 (i.e., $\sum_{i=1}^{N} P_{ij}^{PR} = \sum_{i=1}^{N} \frac{a_{ji}}{K_j^{out}} = 1$). Substituting these values yields:

$$\sum_{i=1}^{N} G_{ij} = \mu(1) + (1-\mu) + \mu(0) = 1$$

Case 2: Node j is a dangling node ($K_j^{out} = 0$)
In this "dead-end" scenario, the node has no outward exposures, meaning $a_{ji} = 0$ for

all i. Consequently, the transition sum $\sum_{i=1}^{N} P_{ij}^{PR} = 0$. However, the Kronecker delta activates, giving $\delta(K_j^{out}, 0) = 1$. The equation resolves as:

$$\sum_{i=1}^{N} G_{ij} = \mu(0) + (1-\mu) + \mu(1) = 1$$

Because the column sum is strictly 1 under all conditions, G is confirmed as a left stochastic matrix. This substitution simplifies the PageRank equation into a pure linear combination:

$$x_i = \sum_{j=1}^{N} G_{ij} x_j \tag{6}$$

In vector form, this precisely yields the eigenvalue equation for $\lambda = 1$:

$$x = Gx \tag{7}$$

Therefore, the PageRank vector x is defined as the principal right eigenvector of the Matrix G associated with the dominant eigenvalue $\lambda_1 = 1$.

3.2 The Generalized MultiRank System

Real-world financial systems are rarely confined to a single type of interaction. To capture the multiplex nature of global banking, where institutions are connected simultaneously through lending, derivatives, and cross-holdings, we transition to the MultiRank algorithm framework, proposed by Rahmede et al. (2018).

3.2.1 Aggregating the Multiplex Layers

Consider a multiplex network with M layers. Each layer α (where $\alpha = 1, 2, \ldots, M$) is represented by its own adjacency matrix $a^{[\alpha]} = (a_{ij}^{[\alpha]})$. The total weight of layer α is defined as the sum of all its edge weights:

$$W^{[\alpha]} = \sum_{i=1}^{N} \sum_{j=1}^{N} a_{ij}^{[\alpha]} \tag{8}$$

To evaluate the relative importance of nodes within a specific layer, we define the normalized in-strength and out-strength:

$$B_{\alpha i}^{in} = \frac{\sum_{j=1}^{N} a_{ji}^{[\alpha]}}{W^{[\alpha]}}, \quad B_{\alpha i}^{out} = \frac{\sum_{j=1}^{N} a_{ij}^{[\alpha]}}{W^{[\alpha]}} \tag{9}$$

The fundamental innovation of MultiRank is the aggregation of these distinct layers into a single "coloured network". Rather than summing the layers uniformly, each layer is scaled by a dynamic layer influence parameter $z^{[\alpha]}$. The elements of the aggregated matrix C are thus defined as:

$$C_{ij} = \sum_{\alpha=1}^{M} a_{ij}^{[\alpha]} z^{[\alpha]} \tag{10}$$

3.2.2 The Non-Linear Coupled Equations

In the aggregated network, the centrality of node i (denoted as x_i) is calculated similarly to PageRank, but operating on the dynamic matrix C:

$$x_i = \mu \sum_{j=1}^{N} \frac{C_{ji}}{K_j} x_j + w v_i \tag{11}$$

where $K_j = \max\left(1, \sum_{i=1}^{N} C_{ji}\right)$ is the generalized out-degree in the coloured network, w is the teleportation scaling factor defined by equation(2), and v_i dictates the teleportation preference.

Similarly, the influence of the layer $z^{[\alpha]}$ is not a static prior, but is determined by the centrality of the nodes active within that layer:

$$z^{[\alpha]} = \frac{1}{\mathcal{N}} \left[W^{[\alpha]}\right]^a \left[\sum_{i=1}^{N} B_{\alpha i}^{in} (x_i)^{s\gamma}\right]^s \tag{12}$$

where $\mathcal{N}$ is a normalization constant ensuring that $\sum_{\alpha=1}^{M} z^{[\alpha]} = 1$.

Remark on the Mathematical Structure: Beyond a Linear Eigensystem
It is mathematically crucial to recognize that Equation (11), unlike standard PageRank, does *not* constitute a linear eigenvalue problem. In standard PageRank ($x = Gx$), the transition matrix G is strictly constant. However, in the MultiRank framework, the matrix elements C_{ji} inherently depend on $z^{[\alpha]}$. As shown in Equation (12), $z^{[\alpha]}$ is a function of the node centralities $(x_i)^{s\gamma}$.

Consequently, the transition matrix itself dynamically shifts as a function of the vector x. Thus, the system described by Equations (11) and (12) is a *non-linear coupled dynamical system*. We cannot simply extract the principal eigenvector using the Perron-Frobenius theorem. Instead, the centralities x and the layers' influences z must be solved through an iterative process. Within this iterative process, the teleportation preference v_i explicitly defines the starting position of random walkers. At iteration step $t = 0$, before any shock propagates through the network, the initial probability mass is strictly distributed according to the teleportation preference:

$$x_i^{(0)} = v_i \quad \text{for } i = 1, 2, \ldots, N$$

Using this initial state, the system updates dynamically for each discrete iteration step $t \geq 0$ through the following sequence:

Step 1: Update Layer Influence. The influence of each layer is determined by the node centralities from the current step t:

$$z^{[\alpha](t)} = \frac{1}{\mathcal{N}} \left[W^{[\alpha]}\right]^a \left[\sum_{i=1}^{N} B_{\alpha i}^{in} \left(x_i^{(t)}\right)^{s\gamma}\right]^s$$

Step 2: Aggregate the Dynamic Matrix. The coloured network matrix is dynamically re-weighted:

$$C_{ij}^{(t)} = \sum_{\alpha=1}^{M} a_{ij}^{[\alpha]} z^{[\alpha](t)}$$

Step 3: Propagate the Random Walk. The centralities for the next step $t + 1$ are computed by coupling the network propagation with the constant teleportation term:

$$x_i^{(t+1)} = \mu \sum_{j=1}^{N} \frac{C_{ji}^{(t)}}{K_j^{(t)}} x_j^{(t)} + w v_i$$

where $K_j^{(t)} = \max\left(1, \sum_{i=1}^{N} C_{ji}^{(t)}\right)$.

This explicit equation shows that the random walker fundamentally originates from the distribution v_i at $t = 0$, and continuously falls back on v_i via the teleportation term wv_i in following steps. The iteration continues until the system reaches a steady-state equilibrium across both variables, formally defined by the convergence criterion $||x^{(t+1)} - x^{(t)}|| < \epsilon$.

3.2.3 Strategic Parameters and Financial Interpretation

The behavior of this non-linear coupling is governed by three critical tuning parameters (a, s, γ):

- **Parameter $a \in \{0, 1\}$:** Determines whether the layer influence scales strictly with its total transaction volume ($a = 1$) or if size bias is neutralized ($a = 0$).
- **Parameter $s \in \{-1, 1\}$:** Defines the "strategic core" property. If $s = 1$, layers gain influence simply by hosting many nodes with high centrality. Crucially, if $s = -1$, it rewards "elite" layers: influence increases when a layer's activity is driven predominantly by a highly concentrated, select group of central institutions, mirroring the mechanics of elite risk-sinks in financial crises.
- **Parameter $\gamma > 0$:** Tunes the intensity of the non-linear contribution of node centralities to the layer influence.

3.3 Theoretical Properties and Syntheses

A robust generalized metric must be able to gracefully recover its foundational algorithm under restrictive boundary conditions. In this section, we provide a rigorous algebraic proof demonstrating that MultiRank is a true generalization of standard PageRank.

Theorem 3.1 (The Collapse to PageRank). *Let a multiplex network be restricted to a single layer ($M = 1$) with adjacency matrix $a_{ij}^{[1]} = a_{ij}$. Under the parameter conditions $a = 0$ and $\gamma = 0$, the MultiRank equations decouple such that the steady-state centrality vector x is given precisely as the solution of the linear eigensystem $x = Gx$, equivalent to the standard PageRank formulation.*

Proof. We begin by evaluating the layer influence equation under the specified boundary conditions. Substituting these into Equation (12) yields:

$$z^{[1]} = \frac{1}{\mathcal{N}} \left[W^{[1]}\right]^0 \left[\sum_{i=1}^{N} B_{1i}^{in} (x_i)^{s(0)}\right]^s = \frac{1}{\mathcal{N}} (1) \left[\sum_{i=1}^{N} B_{1i}^{in} (1)\right]^s \tag{13}$$

To evaluate the summation of the normalized inward weights, we expand B_{1i}^{in} according to its definition:

$$\sum_{i=1}^{N} B_{1i}^{in} = \sum_{i=1}^{N} \frac{\sum_{j=1}^{N} a_{ji}^{[1]}}{W^{[1]}} = \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} a_{ji}^{[1]}}{W^{[1]}} \tag{14}$$

By definition, the double summation of all inward edges is exactly equal to the total weight of the layer $W^{[1]}$. Therefore:

$$\sum_{i=1}^{N} B_{1i}^{in} = \frac{W^{[1]}}{W^{[1]}} = 1 \tag{15}$$

Substituting this back into the layer influence equation gives $z^{[1]} = \frac{1}{\mathcal{N}}(1)^s = \frac{1}{\mathcal{N}}$. Since the normalization constraint requires $\sum_{\alpha=1}^{M} z^{[\alpha]} = 1$, and $M = 1$, it strictly follows that $\mathcal{N} = 1$ and thus $z^{[1]} = 1$.

With the layer influence neutralized and decoupled from the centrality vector x, the system collapses back to the standard PageRank topology through four sequential steps:

Step 1: Matrix Equivalence. The aggregated coloured network matrix simplifies to the standard adjacency matrix:

$$C_{ij} = \sum_{\alpha=1}^{1} a_{ij}^{[\alpha]} z^{[\alpha]} = a_{ij}^{[1]}(1) = a_{ij} \tag{16}$$

Step 2: Out-Strength Alignment. The generalized out-degree naturally simplifies to the standard PageRank out-degree formulation, inherently preserving the mechanism for handling dangling nodes:

$$K_j = \max\left(1, \sum_{i=1}^{N} C_{ji}\right) = \max\left(1, \sum_{i=1}^{N} a_{ji}\right) = k_j^{out} \tag{17}$$

Step 3: Transition Probabilities. As a direct consequence of Steps 1 and 2, the MultiRank random walk transition matrix identically matches the PageRank transition matrix:

$$P_{ij}^{MR} = \frac{C_{ji}}{K_j} = \frac{a_{ji}}{k_j^{out}} = P_{ij}^{PR} \tag{18}$$

Step 4: System Decoupling. By substituting the decoupled transition matrix back into Equation (11), the vector equation becomes

$$x_i = \mu \sum_{j=1}^{N} P_{ij}^{PR} x_j + w v_i \tag{19}$$

To complete the equivalence, we must explicitly address the teleportation scaling factor w. In both frameworks, w is dynamically determined by the network's out-degrees to prevent probability mass from being trapped in dangling nodes. Because Step 2 inherently establishes that the total outgoing exposures align exactly ($\sum_{i=1}^{N} C_{ji} = K_j^{out}$), the MultiRank teleportation factor mathematically reduces to the exact standard PageRank w defined in Equation (2).

Therefore, assuming a uniform teleportation preference vector $v_i = 1$, this equation precisely coincides with the fundamental scalar definition of PageRank (Equation 1). With both the transition matrices and the teleportation components mathematically identical, the unique probability vector solved by the decoupled MultiRank system perfectly coincides with the standard PageRank centrality vector. This completes the proof. $\square$

3.3.1 The Non-Linearity of Layer Aggregation: A Counterexample

Having established that MultiRank elegantly collapses to PageRank under strict boundary conditions, we must now address the inverse question: Is the complex non-linear coupling of MultiRank strictly necessary for multiplex networks? Could we not simply aggregate all layers into a single adjacency matrix $A^{single} = \sum_{\alpha} A^{[\alpha]}$ and apply standard PageRank?

To rigorously prove that systemic centrality is non-additive and that simple aggregation fails to capture true topological risk, we construct a formal counterexample. Let the multiplex network be defined on a common vertex set of $N = 8$ financial institutions, $V = \{A, B, C, D, E, F, G, H\}$. We define two distinct layers, represented by directed graphs $G^{[0]} = (V, E^{[0]})$ and $G^{[1]} = (V, E^{[1]})$ with corresponding adjacency matrices $A^{[0]}$ and $A^{[1]}$:

- **Layer 0 (Routine Transactions):** This layer forms a star-like topology centered at Node A, representing standard, high-frequency reciprocal clearing operations. The directed edge set is formally defined as:

$$E^{[0]} = \{(A, B), (B, A), (A, C), (C, A), (A, D), (D, A), (A, E), (E, A), (A, F), (F, A)\}$$

- **Layer 1 (Risk Sink):** A sparser, highly directional layer representing concentrated risk exposures (e.g., derivatives). The topology structurally forces pathways toward Nodes G and H, acting as absorbing sinks. The directed edge set is:

$$E^{[1]} = \{(A, G), (B, G), (C, G), (G, A), (D, H), (E, H), (F, H), (G, H)\}$$

The Illusion of Single-Layer Aggregation
If we compress this multiplex topology into a single aggregated adjacency matrix $A^{single} = A^{[0]} + A^{[1]}$ and calculate the standard PageRank vector, the algorithm is blinded by the sheer volume of routine connections. Because standard PageRank treats all edges homogeneously, the high degree of Node A in $E^{[0]}$ artificially inflates its centrality (0.2520), making it appear almost as systemically dangerous as the true risk sink, Node H (0.2656). Mathematically, single-layer aggregation creates an illusion of diffused risk, masking the severe underlying vulnerability concentrated in $E^{[1]}$.

The Reality of MultiRank
Applying the generalized MultiRank system to the disaggregated layers yields a drastically different and mathematically superior equilibrium. Because MultiRank's transition matrix is non-linearly coupled with the layer influence $z^{[\alpha]}$, the algorithm endogenously detects that Layer 1 acts as a structural "trap" for random walkers. Consequently, the system assigns a higher dynamic weight to the risk layer. In the converged MultiRank eigenvector, the centrality of Node H experiences a severe non-linear surge (reaching 0.3400), exposing it as the undisputed systemic "black hole" of the network, while the routine hub (Node A) fails to scale.

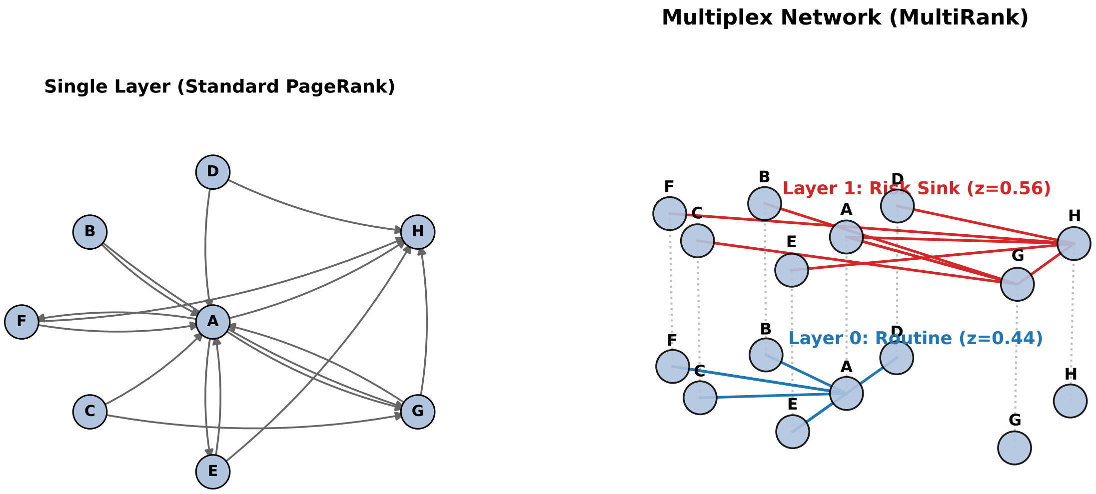


Figure 3: Counterexample demonstrating the non-linearity of layer aggregation. **Left:** Standard PageRank identifies Node A as the core. **Right:** MultiRank exposes Nodes G and H.

Node	**Standard PageRank** (PR)	**MultiRank** (x_i)	**Systemic Shift**
A (Routine Hub)	0.2520	0.2509	Minor Decrease
B	0.0470	0.0264	Decrease
C	0.0470	0.0264	Decrease
D	0.0470	0.0265	Decrease
E	0.1005	0.0918	Minor Decrease
F	0.1005	0.0918	Minor Decrease
G (Risk Sink)	0.1404	0.1465	Minor Increase
H (Risk Sink)	0.2656	**0.3400**	**Severe Surge** ($\uparrow$)

Table 1: Comparison of Node Centralities on the Toy Network. The shift illustrates how MultiRank penalizes routine hubs (Node A) while exposing true risk sinks (Nodes G and H).

Conclusion of the Counterexample

This stark divergence mathematically proves that $PR(\sum A^{[\alpha]}) \neq MR(\{A^{[\alpha]}\})$. The propagation of financial shocks across different asset classes is fundamentally non-linear. Compressing multiplex networks into a single matrix prematurely homogenizes different risk vectors, leading to dangerous misidentifications of the systemic hubs. Therefore, the coupled, non-linear architecture of MultiRank is not merely an optional extension, but a mathematical necessity for accurate systemic risk assessment.

4 Data and Empirical Network Topology

Transitioning from theoretical models to empirical reality, this chapter outlines the construction of a real-world multiplex financial network. We map the topological shifts of the global banking system before, during, and after the 2008 Global Financial Crisis.

4.1 Data Source and Empirical Scope

To construct an accurate representation of the global financial architecture, we utilize the Locational Banking Statistics (LBS) provided by the Bank for International Settlements (BIS) [2]. The LBS database systematically captures outstanding claims and liabilities of internationally active banks, serving as the standard for mapping cross-border interconnectedness.

In our empirical topology, the set of nodes $V = \{1, 2, \dots, N\}$ represents the national banking sectors. A weighted directed edge from country i to country j indicates that i holds a financial claim against j (measured in millions of USD). Crucially, this edge direction naturally maps the risk contagion pathway: if country j defaults, country i suffers a direct capital loss. This topology excludes domestic credit exposures. Consequently, the resulting MultiRank centrality strictly quantifies the risk of cross-border contagion.

4.2 Multiplex Network Construction

To rigorously capture the heterogeneous nature of financial exposures, we construct a two-layer multiplex network ($M = 2$). We map specific instrument classifications from the BIS LBS dataset to distinct adjacency matrices:

- **Layer 0 ($A^{[0]}$, Total Claims):** Constructed using the "All instruments" classification. This aggregated matrix captures the baseline topological structure, encompassing loans, debt securities, and other derivatives.
- **Layer 1 ($A^{[1]}$, Loans and Deposits):** Constructed using the restricted "Loans and deposits" classification. This matrix explicitly isolates the highly liquid interbank funding channels.

4.3 Timeline Selection: The Crisis Arc

To isolate contagion dynamics from ongoing policy interventions, we analyze three discrete quarterly snapshots of the global financial network, defined here as the "Crisis Arc":

- **2007 Q2 (Pre-Crisis Baseline):** Captures the network topology prior to the subprime mortgage disruption, serving as a baseline of maximum structural connectivity and liquidity.
- **2008 Q4 (Crisis Peak):** Captures the topology immediately following the Lehman Brothers default. This snapshot isolates the precise moment of structural fracture within the interbank lending market (Layer 1).
- **2009 Q4 (Post-Crisis Phase):** Captures the network post-deleveraging and central bank intervention (e.g., Quantitative Easing), allowing us to measure the topological regime shift and the reallocation of systemic risk.

By analyzing these topological snapshots, we can observe how shifts in MultiRank centrality result from the evolving structure of the global financial network.

4.4 Empirical Topological Snapshot: The 2008 Crisis Peak

Figure 4 displays the global financial network at the peak of the 2008 Q4 crisis. Our full simulation models 43 sovereign nodes. However, to maintain visual clarity, this figure plots only the top 25 national banking sectors (ranked by total weighted degree) and their top three directed outgoing exposures. The specific countries included in this visualization are labeled directly on the respective nodes in the graph.

Figure 4 shows the two-layer structure defined in Section 4.2. Vertical dotted lines connect the same country across both layers. Layer 0 (bottom) represents total claims, while Layer 1 (top) represents loans and deposits. This layout illustrates how different asset types are linked within each national banking system.

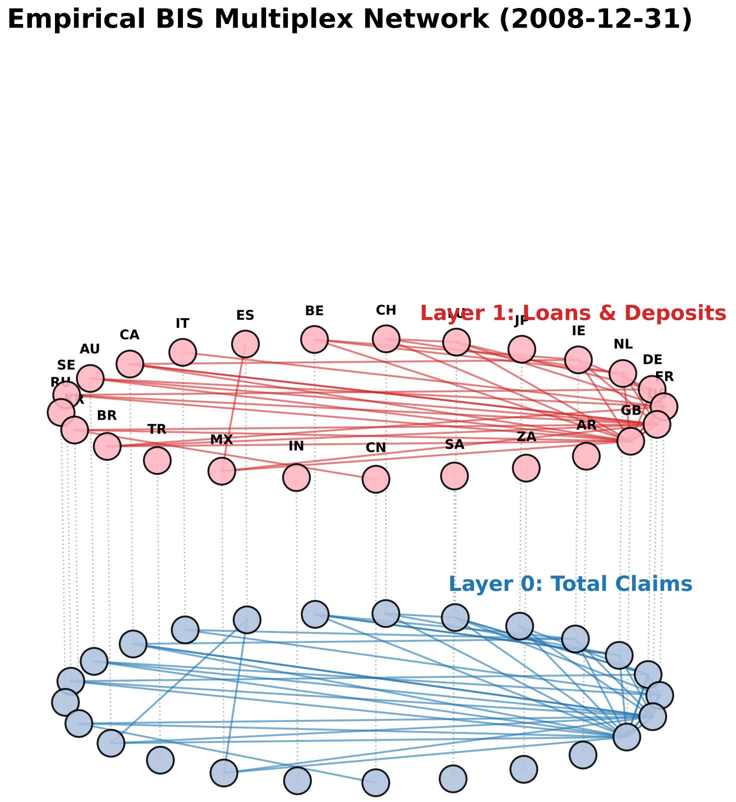


Figure 4: Empirical Multiplex Topology of the Global Banking Network (2008 Q4)

Notes: This structural snapshot captures the topological filtration of the top 25 core national banking sectors at the peak of the 2008 financial crisis. Nodes are arranged in a unified circular layout. **Bottom (Layer 0, Blue):** Total claims representing broad systemic exposure. **Top (Layer 1, Red):** Loans and deposits representing highly liquid, volatile funding channels. Directed edges map the primary outward risk exposures (restricted to the top 3 per node to highlight the primary contagion vectors).

5 Contagion Dynamics

To simulate the propagation of financial distress across the global banking network, we employ a discrete-time deterministic threshold model adapted from the framework proposed by del Rio-Chanona et al. [4]. This chapter mathematically defines the mechanics of systemic collapse, modeling how localized defaults cascade through both structural exposures and wholesale funding channels.

5.1 The Solvency Shock Mechanism

Following the framework of del Rio-Chanona et al. [4], we establish a discrete-time deterministic threshold model where financial distress can propagate horizontally (within a specific asset layer) and vertically (in all layers of a country).

1. Network State and Assets

In a multiplex network, a country might default in one specific asset class while remaining solvent in another. Therefore, in any discrete time step t, the state of country i is defined specifically for each layer α using a binary variable:

$$S_{i\alpha}(t) \in \{0, 1\} \quad (0 : \text{Healthy}, 1 : \text{Distressed}) \tag{20}$$

The asset of a country within a specific layer α is the sum of its outgoing claims directed in that layer:

$$A_{i\alpha} = \sum_{j=1}^{N} A_{ij}^{[\alpha]} \tag{21}$$

However, the BIS dataset only maps specific cross-border exposures, not the entire domestic economy. To calculate a country's actual total assets (A_i^{total}), we sum its exposures across all M layers and scale the result by an external asset ratio θ:

$$A_i^{\text{total}} = \frac{1}{\theta} \sum_{\alpha=1}^{M} \sum_{j=1}^{N} A_{ij}^{[\alpha]} \tag{22}$$

2. Loss Propagation

Contagion occurs when borrower nations default, causing direct credit write-downs. The loss experienced by country i in layer α at time t is the exact sum of its directed exposures to failed partners in that specific layer:

$$L_{i\alpha}(t) = \sum_{j=1}^{N} A_{ij}^{[\alpha]} S_{j\alpha}(t) \tag{23}$$

The total accumulated loss across all financial layers is simply $L_i^{\text{total}}(t) = \sum_{\alpha=1}^{M} L_{i\alpha}(t)$.

3. The Dual-Threshold Rule

The resilience of a country depends on two capital buffers: a horizontal buffer (b_i) for individual layers and a vertical buffer (c_i) for its total assets. If a country breaches the limit in any single layer or if its combined total losses exceed the vertical buffer, it enters complete insolvency. In this model, a default in any specific part of the network triggers a total default across all layers. The deterministic state update is:

$$S_{i\alpha}(t+1) = \begin{cases} 1 & \text{if } \frac{L_{i\alpha}(t)}{A_{i\alpha}} > b_i \quad \text{(Layer Default)} \\ 1 & \text{if } \frac{L_i^{\text{total}}(t)}{A_i^{\text{total}}} > c_i \quad \text{(Systemic Default)} \\ S_{i\alpha}(t) & \text{otherwise} \end{cases} \tag{24}$$

This irreversible rule ensures that structural shocks propagate in a monotonic manner, spreading through inter-layer counterparty linkages as well as through risk transfers across layers.

5.2 The Multiplex Shock: Adding Liquidity Contagion

The 2008 crisis demonstrated that liquidity hoarding can trigger systemic collapse faster than traditional solvency shocks. During a panic, institutions withdraw wholesale funding to protect their own liquidity. This ”flight to safety” can cause a healthy node to default simply because its short-term funding is suddenly removed.

Applying the Loss Function

We apply the general model from Section 5.1 to the two-layer network defined in Section 4.2. For any country i, the total multiplex loss $L_i^{\text{total}}(t)$ is the sum of asset write-downs and funding withdrawals:

$$L_i^{\text{total}}(t) = L_{i0}(t) + \rho L_{i1}(t) \tag{25}$$

- $L_{i0}(t) = \sum_{j=1}^{N} A_{ij}^{[0]} S_{j0}(t)$ is the loss from defaults in Layer 0 (Total Claims).
- $L_{i1}(t) = \sum_{j=1}^{N} A_{ij}^{[1]} S_{j1}(t)$ is the funding withdrawn by distressed partners in Layer 1 (Loans and Deposits).
- $\rho \in [0, 1]$ represents the severity of the credit crunch, or the fraction of withdrawn funding that cannot be replaced.

Systemic Default Condition

The default logic remains consistent with the dual-threshold rule in Equation (24). A country enters complete insolvency if its total combined loss exceeds the vertical capital buffer c_i:

$$\text{Systemic Default if: } \frac{L_i^{\text{total}}(t)}{A_i^{\text{total}}} > c_i \tag{26}$$

Incorporating $\rho L_{i1}(t)$, this mechanism captures how a liquidity freeze accelerates systemic collapse by breaching capital buffers more rapidly than solvency losses alone.

5.3 Parameter: Heterogeneous Capital Buffers

We assign different capital buffers to countries to reflect variations in leverage across the global economy. Central financial hubs typically operate with higher leverage and smaller capital reserves than peripheral nations. Consequently, we determine each country's buffer size based on its importance in the network: the more central a country is, the smaller its assigned capital buffer.

We classify the nodes into two distinct risk profiles:

- **Core Hubs (G-SIBs):** Identified purely by their apex MultiRank centrality, these highly interconnected institutions operate with higher leverage to maximize liquidity velocity. To reflect this fragility, we assign these core nodes strict heuristic bounds of $b_i = 0.03$ (3%) for the horizontal buffer and $c_i = 0.05$ (5%) for the vertical buffer.
- **Periphery (Rest of the World):** The less connected, peripheral nations generally operate with more conservative leverage. We fortify these nodes with higher heuristic bounds of $b_i = 0.08$ (8%) and $c_i = 0.12$ (12%).

The capital buffers assigned are theoretical parameters, not empirical data. This simplified setup serves as a stress test to strictly isolate the network's structural fragility under liquidity constraints.

6 Simulation Results and Systemic Collapse

Having established the MultiRank mathematical framework and the dual-channel contagion mechanics, we now deploy the model on the empirical BIS dataset. This chapter visualizes the structural evolution of the global financial network across the "Crisis Arc" and simulates the non-linear propagation of systemic defaults.

6.1 Static Topological Analysis: Unmasking the Risk Sinks

Before simulating dynamic contagion, we examine the static topological equilibrium of the network. Prior to dynamic simulation, we examine the static network equilibrium across 2007 Q2, 2008 Q4, and 2009 Q4. This comparison identifies the structural shift in systemic risk during the crisis arc. Full centrality scores are provided in Appendix 2.

The Macro Regime Shift in Layer Influence. Figure 5 tracks the evolution of the layer influence factor ($z^{[\alpha]}$). Throughout the observed period, Layer 0 (Total Claims) consistently maintains a higher structural influence than Layer 1 (Loans and Deposits).

Rather than a complete reversal, the crisis peak in 2008 Q4 is characterized by a severe widening of the influence gap. The data indicates that $z^{[0]}$ increases from its 2007 Q2 baseline of 0.6245 to a peak of 0.6424. Simultaneously, $z^{[1]}$ drops from 0.3755 to a systemic minimum of 0.3576. Following this peak, the network exhibits a V-shaped recovery trend by 2009 Q4, with $z^{[1]}$ rebounding to 0.3640 and $z^{[0]}$ contracting to 0.6360.

This V-shaped trajectory explicitly maps the mechanics of the credit crunch. During the crisis, institutions actively hoarded liquidity by withdrawing short-term cross-border loans, which mathematically reduced the connectivity and topological weight of Layer 1. In contrast, the broader exposures in Layer 0—such as long-term debt and complex derivatives—remained illiquid and trapped in the balance sheets, temporarily inflating their relative dominance in the MultiRank equilibrium.

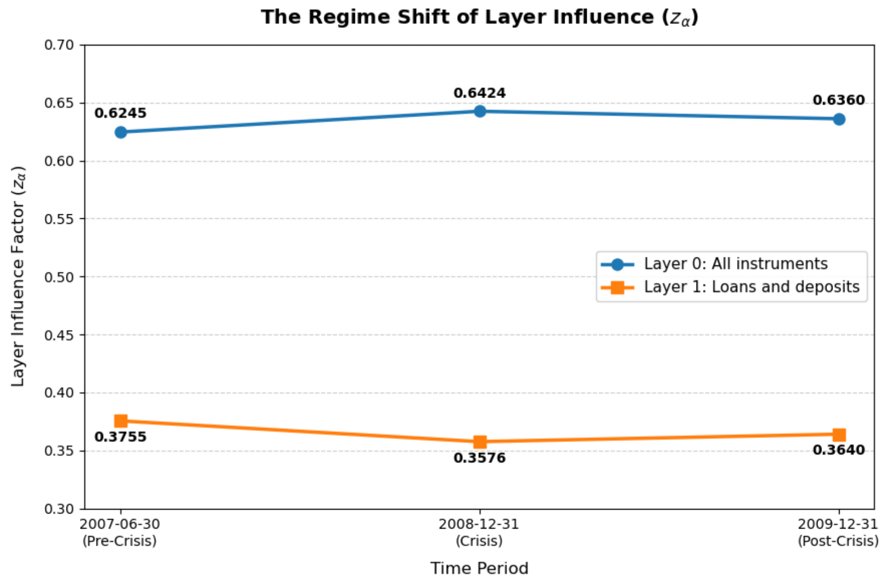


Figure 5: The regime shift of layer influence

Micro Consolidation: Rank Stickiness and Centrality Diffusion

Despite the macroscopic shift in layer influence, the ordinal hierarchy of the top 10 sovereign nodes remains persistently stable (Figure 6). The United Kingdom and the United States consistently occupy the apex of the network, with only marginal rank variations occurring among other core European hubs (e.g., FR, DE, NL).

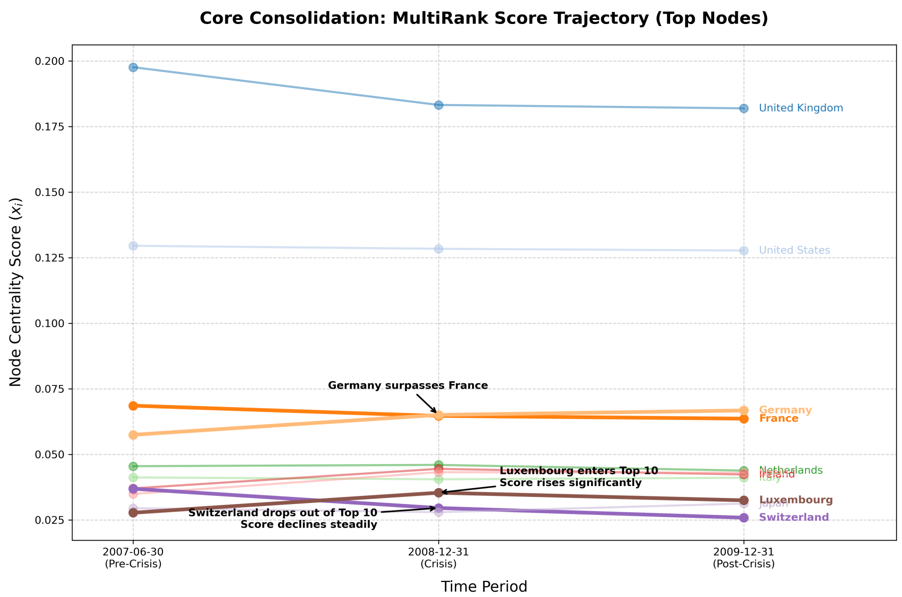


Figure 6: Rank Evolution

However, this rank stability obscures a fundamental reallocation of absolute financial weight. Figure 7 details the underlying reality through a heatmap of the absolute Multi-Rank centrality scores (x_i). Across the crisis arc, the absolute centrality of the dominant nodes strictly contracts. For instance, from 2007 Q2 to 2009 Q4, the centrality score of GB drops from 0.1976 to 0.1819, and the US declines from 0.1295 to 0.1277.

Because network centrality distributes relative importance within a closed topological system, this erosion at the absolute core mathematically necessitates a structural rise elsewhere. This diffusion of centrality reflects the post-crisis macroeconomic deleveraging of traditional transatlantic hubs and the simultaneous growth of emerging and Asian financial centers. MultiRank quantifies this redistribution prior to any disruption in the nominal ranking order.

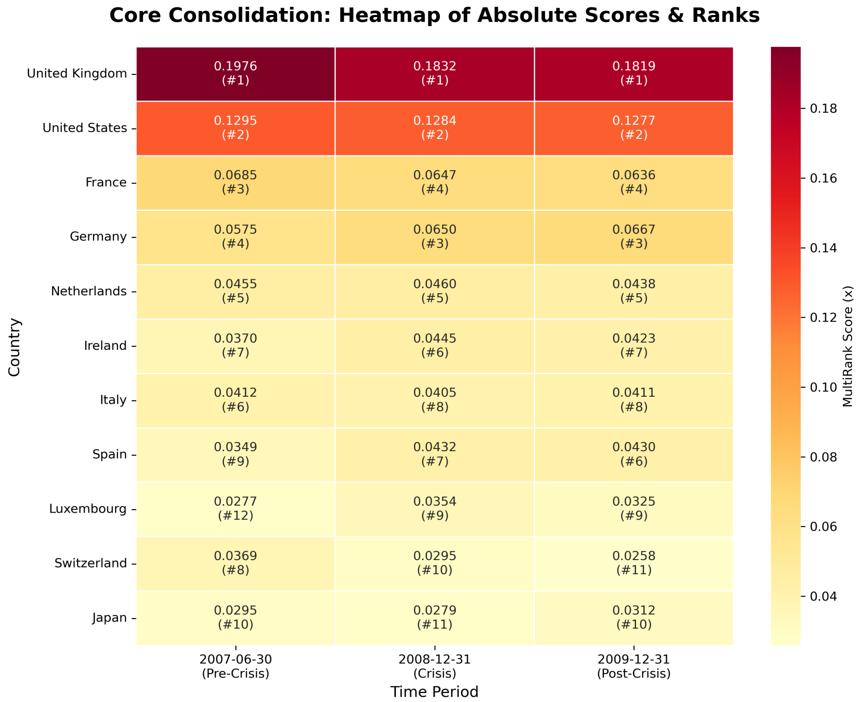


Figure 7: Micro Consolidation

In summary, static analysis identifies two structural conditions: a contraction in short-term liquidity channels and a gradual redistribution of global financial weight. However, static centrality scores only map underlying vulnerabilities; they cannot capture the non-linear mechanics of contagion. To model how a localized default propagates through the network, the following sections transition from static observation to dynamic stress testing.

6.2 Identifying Systemic Fragility: The V-Shaped Contagion Trend

To evaluate the baseline vulnerability of the global banking network, we first simulate a pure solvency shock under a uniform capital buffer assumption. We apply identical initial shocks to nodes across the multiplex networks of three distinct periods: 2007 (pre-crisis), 2008 Q4 (peak crisis), and 2009 Q4 (post-crisis). Figure 8 illustrates the number of nodes that survive after the cascade in each period.

As clearly illustrated in Figure 8, the simulation reveals a distinct V-shaped trend in network survivability. Across multiple diverse shock scenarios—most notably when the contagion originates in Italy (IT), the United States (US), Spain (ES) or Switzerland (CH)—the number of surviving nodes experiences a drastic drop to its absolute minimum in 2008 Q4, before exhibiting a structural recovery in 2009 Q4.

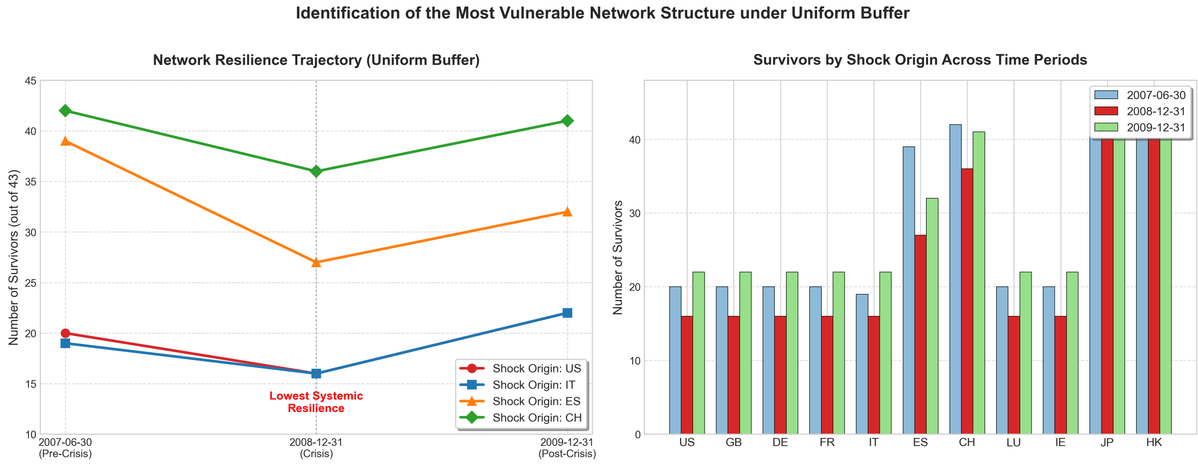


Figure 8: Network resilience under a uniform buffer across different shock origins and time periods.

Topologically, the V-shaped trajectory indicates that the 2008 Q4 network possessed the most fragile structural configuration. During this period, the specific distribution of the exposures of the internodes and the cross-layer links maximized the efficiency of the contagion.

However, while the uniform baseline simulation identifies this peak topological fragility, it remains a theoretical simplification. During the 2008 crisis, capital distribution was highly heterogeneous. The central core hubs operated with higher leverage and lower effective capital buffers compared to peripheral nodes.

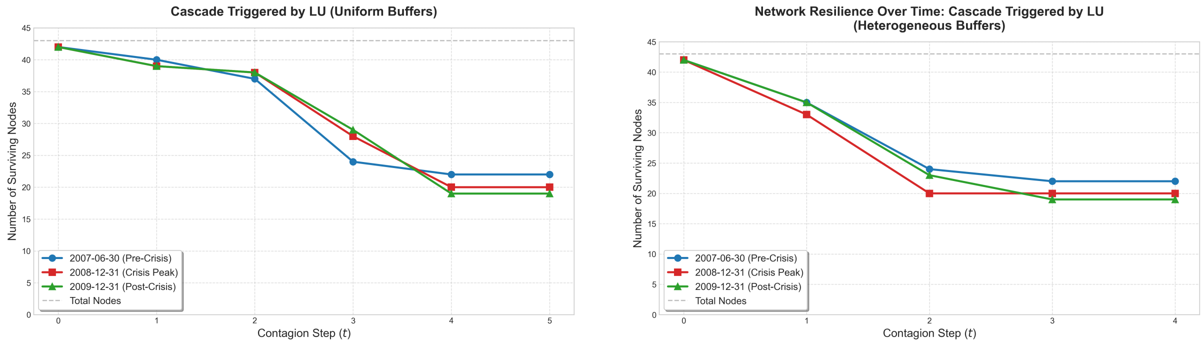
Consequently, applying a uniform buffer obscures the specific structural vulnerabilities of the core network. To more accurately model the 2008 crisis, the following section re-evaluates the 2008 Q4 network using a heterogeneous capital buffer model, where a node's capital buffer is inversely proportional to its topological centrality.

6.3 Heterogeneous Capital Buffers: The Vulnerability of the Core

To go beyond the limitations of the uniform buffer assumption discussed in Section 6.2, we introduce a heterogeneous capital buffer distribution. This approach more accurately reflects the empirical reality of the 2008 Q4 network, where highly leveraged institutions had already depleted much of their capital holding toxic assets.

We determine the buffer allocation using the topological hierarchy established earlier. Specifically, we designate the top 10 nodes with the highest MultiRank scores as the "Core" of the network. Remarkably, this purely data-driven topological classification closely aligns with the Bank for International Settlements' (BIS) real-world framework for identifying Global Systemically Important Banks (G-SIBs) and their primary jurisdictions. To model their high leverage during the crisis peak, we assign a significantly lower capital buffer ($c_i = 0.05$ (5%) $b_i = 0.03$ (3%)) to these Core nodes, while the remaining Periphery's nodes receive a standard, higher buffer ($c_i = 0.12$ (12%) $b_i = 0.08$ (8%)).

For the stress test, we select Luxembourg—an underlying European financial hub—as the initial defaulting node, rather than the United States. Although the US holds the highest MultiRank centrality, its outsized financial mass causes a direct shock to trigger an immediate, systemic collapse. This rapid system-wide failure obscures the underlying propagation mechanics of the contagion. Initiating the shock from Luxembourg provides a perturbation of sufficient magnitude to stress the network while remaining bounded enough to allow observation of the contagion pathways through specific topological vulnerabilities.



(a) Uniform buffer (b) Heterogeneous buffer

Figure 9: Comparison of network survival rates over time.

Figure 9 compares network survival under uniform and heterogeneous buffers. To clarify temporal mechanics, we define a contagion step t as a single discrete iteration of loss propagation and default threshold evaluation. Under heterogeneous buffers, the survival curve drops more steeply. The initial shock from Luxembourg propagates immediately to the core hubs. Constrained by their lower capital buffers, these central nodes reach their default thresholds earlier in the discrete timeline.

Figure 10 maps the step-by-step structural progression of this cascade. The data confirm that the contagion does not diffuse uniformly. Instead, the risk pathways are strictly channeled through the core nodes. Upon default, these high-MultiRank hubs act as systemic amplifiers, propagating the cascade outward to the periphery. This core-failure mechanism drives the lower final survival state compared to the uniform baseline.

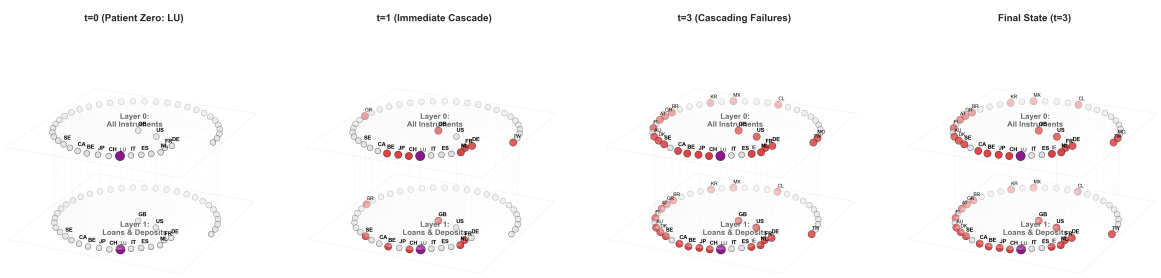


Figure 10: visualization of the contagion cascade under heterogeneous buffers.

6.4 Dual-Channel Contagion: The Credit Crunch and Systemic Annihilation

Expanding the framework to include a liquidity contagion channel introduces a destabilizing feedback loop (Figure 11). An initial solvency shock in Layer 0 triggers defensive liquidity hoarding in Layer 1, as surviving institutions withdraw cross-border funding. This credit contraction inflicts secondary funding shocks on borrower nodes. The aggregated stress from both asset write-downs and funding drains rapidly breaches the lower capital buffers of the core hubs, initiating a self-reinforcing default cascade.

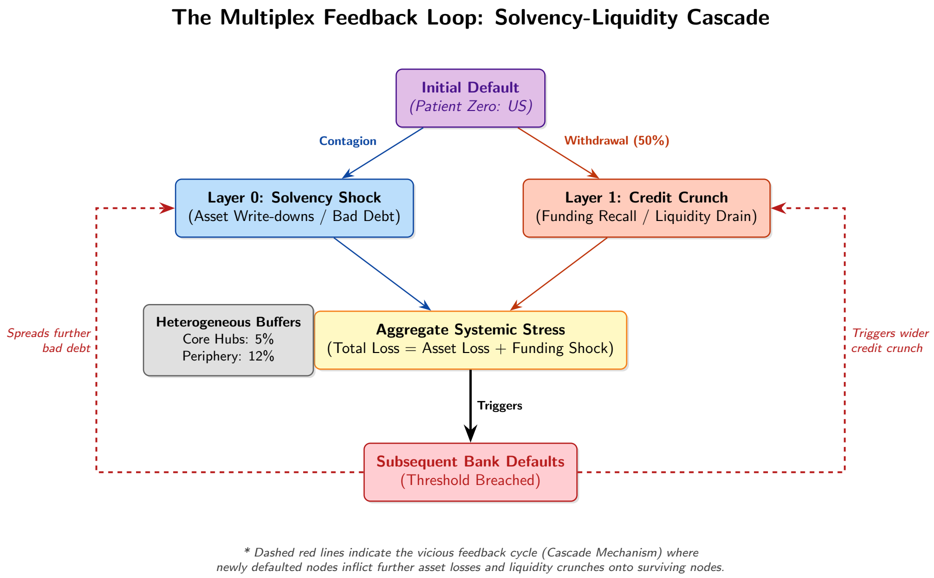


Figure 11: The multiplex feedback loop. An initial solvency shock in Layer 0 triggers a credit contraction in Layer 1, combining to breach the heterogeneous buffers of surviving nodes and propagate the contagion.

To quantify the impact of this multiplex feedback loop, we simulate shocks originating from distinct G-SIB hubs under varying severities of credit contraction, governed by the liquidity parameter $\rho \in \{0, 0.3, 0.5\}$ as defined in Equation (25). Figure 12 (left) presents the survival curves following an initial default of the United States. Driven by its MultiRank centrality, a US default precipitates an immediate global liquidity drain. Consequently, the global banking network collapses to a near-zero survival state within the first iteration step ($t = 1$), strictly independent of the magnitude of ρ. The sheer volume of US outward exposures immediately overwhelms the heterogeneous capital buffers (b_i, c_i) of neighboring nodes.

To observe the cascade mechanics with greater granularity, Figure 12 (right) illustrates a contagion initiated by Germany. This scenario exhibits a distinct temporal dynamic strictly dependent on the credit crunch parameter. Specifically, under a moderate liquidity contraction ($\rho = 0.3$), the network demonstrates temporary structural resilience at $t = 1$, as the initial localized solvency losses (L_{i0}) fail to immediately breach the systemic default thresholds defined in Equation (26). However, as the multiplex feedback loop activates, the secondary liquidity drain (ρL_{i1}) compounds the aggregate systemic stress. By $t = 2$, this combined loss vector strictly breaches the remaining capital buffers, driving the system to complete collapse.

The dual-channel simulations establish that the unhindered multiplex network strictly converges to a terminal equilibrium of systemic collapse. Because total failure is the baseline mathematical outcome, the research objective transitions from evaluating structural fragility to optimizing policy intervention. Constrained by finite regulatory capital, this presents a distinct allocation problem: should liquidity be injected into the initial shock

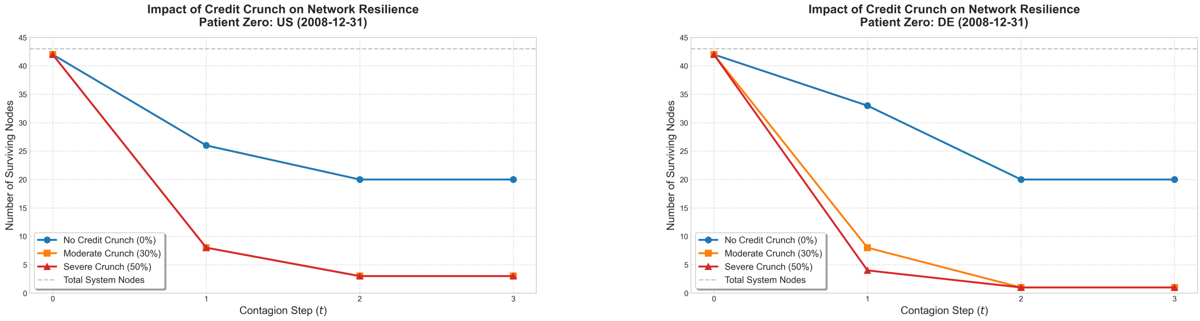


Figure 12: Network survival rates under dual-channel contagion. (Left) A shock originating from the US causes rapid global collapse across all contraction levels. (Right) A shock originating from Germany shows a delayed collapse, notably surviving one initial step under a 30% credit contraction before total systemic failure.

origin to neutralize the trigger, or directed toward the high-MultiRank core hubs to firebreak the contagion? The following section evaluates the systemic return on investment (ROI) of these targeted strategies.

7 Policy Intervention: The ROI of Bailouts in Multiplex Networks

Our dual-channel simulations show that an unmitigated credit crunch causes a total network collapse. Therefore, external policy intervention is required to halt the cascade. To properly evaluate these interventions, we must shift our primary analytical metric (Figure 13).

7.1 Motivation and Metric Shift: From Nodes to Economic Mass

Traditional network science measures resilience by counting the absolute number of surviving nodes. As shown in the “Topology View” (Figure 13, left), this metric assumes all nodes are equal. While suitable for homogeneous networks, it fails when applied to the global financial system.

Financial networks are highly skewed. A peripheral bank and a Global Systemically Important Bank (G-SIB) hold vastly different economic masses. The “Economic Mass View” (Figure 13, right) reflects this reality: a single core hub often contains the majority of the network’s financial volume. Consequently, a 90% topological loss (e.g., losing 9 out of 10 peripheral nodes) might only result in a 10% loss of total economic mass, provided the core hub is firewalled.

Policymakers operate with finite capital and cannot save every institution. Interventions must therefore optimize the systemic return on investment (ROI). Because counting nodes obscures the true macroeconomic impact of a crisis, we transition our primary evaluation metric from the *Number of Surviving Nodes* to the *Asset Preserved Percentage*. The following sections use this metric to evaluate bailout strategies, comparing interventions targeted at the initial shock source versus those targeted at the core hubs.

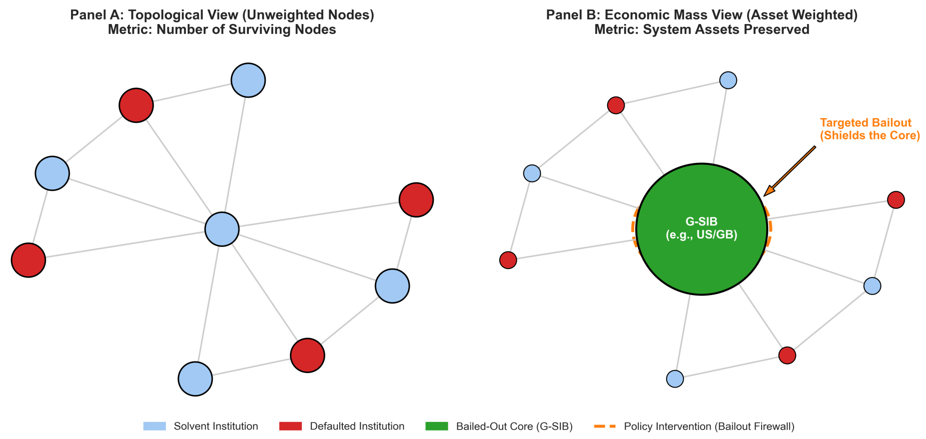


Figure 13: Comparison between topological and economic mass evaluation. In a heterogeneous financial network (right), securing a core hub preserves most of the economic mass, even if a large number of peripheral nodes default.

7.2 Theoretical Benchmark: Targeted Intervention at the Shock Origin

To establish a theoretical benchmark, we simulate a targeted intervention at the initial shock source. Injecting external liquidity directly into the originating node neutralizes the solvency shock before the multiplex feedback loop activates.

As shown in Figure 14, this strategy yields 100% network preservation. Regardless of whether the crisis originates in the United States, Germany, or Luxembourg, the targeted rescue reduces node defaults to zero and maximizes asset preservation. Averting the initial shock prevents Layer 1 liquidity hoarding and completely halts the contagion cascade.

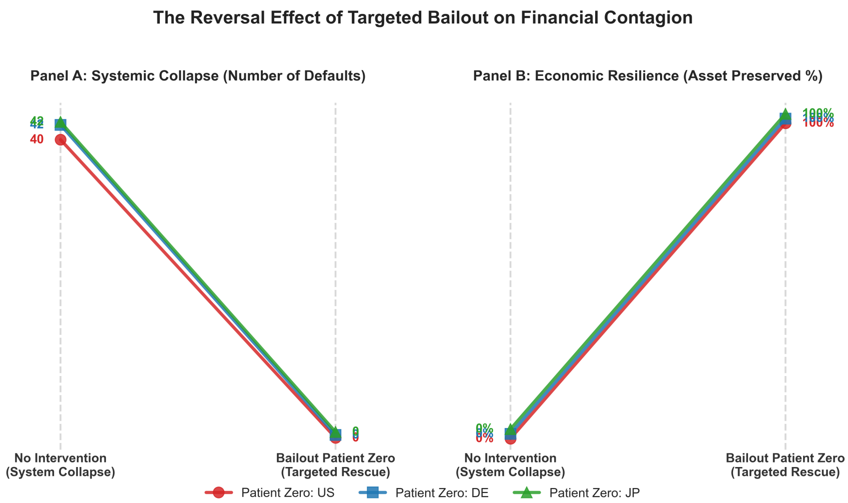


Figure 14: Comparison of unmitigated contagion versus targeted rescue. Shielding the initial shock source strictly preserves both network topology and total systemic assets.

While this targeted rescue establishes an optimal upper bound for intervention efficiency, it is constrained by two structural limitations:

1. **Information Asymmetry:** Real-time interbank exposures lack transparency. Iden-

tifying the exact source of distress before contagion breaches neighboring capital thresholds is practically difficult during the early stages of a crisis.

2. **Jurisdictional Boundaries:** The financial network is global, but regulatory capital is domestic. Central banks operate under strict domestic mandates and generally cannot deploy capital to bail out foreign institutions (e.g., the US Federal Reserve directly intervening in Luxembourg).

Due to these informational and jurisdictional constraints, preemptive global bailouts are rarely feasible. Policymakers are therefore forced to adopt defensive, sub-optimal strategies. Consequently, the analytical focus shifts: given finite resources, how can authorities optimally allocate liquidity to shield the core hubs?

7.3 The Intervention Landscape: Universal Firewalls and Regional Dependencies

To evaluate core firefighting strategies, we simulate crisis origins across five geographical profiles: the US, GB, DE, JP, and AU. Figure 15 plots the systemic ROI (Asset Preserved Percentage) against node centrality for various bailout targets. Numerical results are detailed in Table 6.

Excluding the unfeasible source bailout, the highest consistent preservation across all scenarios is achieved by rescuing the top MultiRank hubs (GB, FR, and the US). Regardless of the shock's origin, these high-centrality nodes act as universal structural firewalls, reliably preserving a baseline of the network's economic mass.

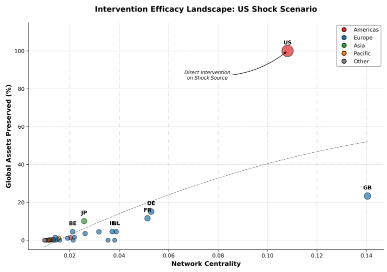
However, the AU shock scenario (Figure 15e) reveals a critical localized anomaly: targeted interventions in Italy (IT) and Spain (ES) yield disproportionately high ROI despite their lower global centrality.

This anomaly explicitly answers why a multiplex framework is mathematically necessary. If exposures were compressed into a traditional single-layer network, the sheer aggregate volume of the US and GB would computationally bias all risk propagation toward the global core, obscuring specific vulnerabilities. By disaggregating the network, the multiplex model captures inter-layer mechanics: a localized solvency shock (Layer 0) in AU triggers specific, highly concentrated liquidity withdrawals (Layer 1) from IT and ES. The multiplex topology isolates these direct contagion corridors that bypass the primary hubs, enabling policymakers to identify highly efficient, crisis-specific regional firebreaks.

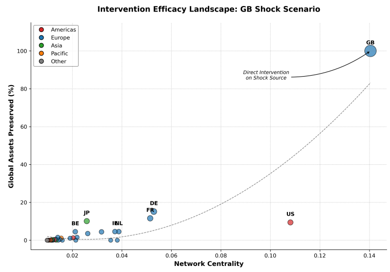
7.4 Concluding Remarks on Simulation Results

The simulations establish that systemic failure in the 2008 multiplex network is primarily driven by the coupling of solvency and liquidity channels. While an isolated solvency shock mostly causes localized peripheral defaults, the secondary credit contraction acts as the true contagion accelerator. This liquidity drain rapidly depletes the lower capital buffers of high-centrality hubs, converting localized distress into a global cascade.

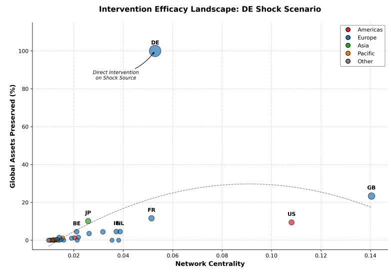
Consequently, this framework optimizes resource allocation during a liquidity crisis. Under severe capital constraints, immunizing the entire network is mathematically impossible. To maximize preserved economic mass, regulatory capital must strictly target high-MultiRank core institutions. Firewalling these topological hubs effectively breaks the multiplex feedback loop.



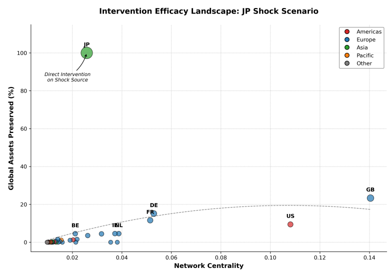
(a) US Shock



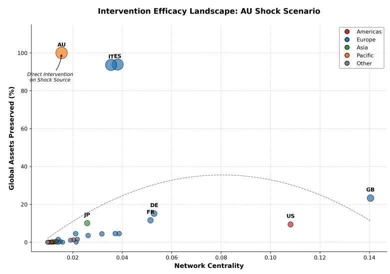
(b) GB Shock



(c) DE Shock



(d) JP Shock



(e) AU Shock

Figure 15: Intervention Landscapes across diverse crisis origins. While rescuing the shock source guarantees 100% preservation, secondary interventions show GB, FR, and the US act as universal firewalls. The AU shock uniquely exposes multiplex contagion corridors directing risk toward IT and ES.

8 Discussion and Limitations

While the dual-channel multiplex model provides a clearer framework for systemic risk, it contains several methodological limitations that guide future research.

First, the analysis relies on a static network snapshots (2008 Q4, 2007 Q4, 2009 Q4). This approach isolates vulnerabilities at a specific moment but excludes dynamic network rewiring. In reality, institutions continuously adjust exposures and cut credit lines during a crisis. Future models should utilize time-varying adjacency matrices to capture these structural adaptations.

Second, the contagion model is deterministic, applying a uniform default threshold ($\theta = 0.3/0.5$) across all nodes. This computational simplification omits the heterogeneous risk appetites of real financial institutions. Future research could introduce stochastic variables to model asymmetric information and market behavior.

Third, the capital buffers (b_i, c_i) are assigned heuristically based on topological centrality. While this isolates structural effects, it remains a theoretical abstraction. A sovereign node's actual defensive capacity depends on specific metrics, such as Basel Tier 1 capital ratios. Future empirical work must calibrate these thresholds using historical accounting data.

Finally, simulations indicate that the network experiences a sudden, non-linear shift from partial survival to complete collapse under severe shocks. Because this thesis focused on mapping the contagion cascade, we did not analytically calculate the exact mathematical boundary of this failure. Identifying the precise critical threshold that triggers systemic collapse is a necessary next step for developing macro-prudential early warning systems.

8.1 Policy Implications: Topology-Based Capital Allocation

Current regulatory frameworks typically assign capital buffers based on a bank's individual asset size. However, our simulations demonstrate that systemic vulnerability is a network property. In a multiplex system, an institution's capacity to propagate risk depends strictly on its topological position across different asset layers, rather than its total asset volume alone.

Therefore, resource allocation can be mathematically optimized using network metrics. Distributing capital buffers based on layer-specific centralities, such as MultiRank, provides a strictly more efficient alternative to size-based rules. By directly targeting the topological core, authorities can construct effective structural firewalls to maximize the preservation of global economic mass using limited regulatory capital.

9 Conclusion and Future Work

This thesis models systemic financial risk using multiplex network theory. By combining a dual-channel contagion framework with the MultiRank algorithm, the simulations demonstrate that secondary liquidity contractions drive systemic collapse more severely than isolated solvency shocks.

Furthermore, the intervention analysis indicates that under severe capital constraints, targeting topological core hubs preserves a strictly higher percentage of total network assets compared to uniform or purely size-based bailouts.

However, this model is constrained by its reliance on a static network snapshot and deterministic default thresholds, which omit the stochastic dynamics of real markets. Building upon these simulations, future research should focus on analytically deriving the exact mathematical critical points that trigger non-linear systemic failure.

A Full Simulation Data Tables

Table 3: Complete Uniform Contagion Results Across 2007–2009 (Corrected)

Category	**Victim (Shock Origin)**	**Buffers (L/T)**	**Surviving Nodes ($N = 43$)**		
			2007 Q4	**2008 Q4**	**2009 Q4**
Global	US	0.10 / 0.15	22	20	19
Global	GB	0.10 / 0.15	22	20	19
EU Core	DE	0.10 / 0.15	22	20	19
EU Core	FR	0.10 / 0.15	22	20	19
EU	IT	0.10 / 0.15	21	19	18
EU	ES	0.10 / 0.15	41	39	40
Hub	CH	0.10 / 0.15	42	42	42
Hub	LU	0.10 / 0.15	22	20	19
Hub	IE	0.10 / 0.15	22	20	19
Asia	JP	0.10 / 0.15	42	41	41
Asia	HK	0.10 / 0.15	42	42	42
Asia	KR	0.10 / 0.15	42	42	42
Control	CA	0.10 / 0.15	42	42	42
Control	AU	0.10 / 0.15	42	42	41
Control	NZ	0.10 / 0.15	41	42	40
Control	IL	0.10 / 0.15	42	42	42
Systemic	US + GB	0.10 / 0.15	22	20	19
Systemic	JP + HK + KR	0.10 / 0.15	40	39	39
Systemic	IT + ES + IE	0.10 / 0.15	20	18	17

Note: All institutions share a uniform Layer/Total buffer of 0.10/0.15. Network size $N = 43$.

Table 4: Complete Heterogeneous Contagion Results Across 2007–2009

Category	**Victim (Shock Origin)**	**Surviving Nodes ($N = 43$)**		
		2007 Q1	**2008 Q4**	**2009 Q4**
Global	US	22	20	19
Global	GB	22	20	19
EU Core	DE	22	20	19
EU Core	FR	22	20	19
EU	IT	21	19	18
EU	ES	21	19	18
Hub	CH	22	20	19
Hub	LU	22	20	19

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Table 4 – continued from previous page

Category	Victim (Shock Origin)	Surviving Nodes ($N = 43$)		
		2007 Q1	**2008 Q4**	**2009 Q4**
Hub	IE	22	20	19
Asia	JP	22	20	19
Asia	HK	42	42	42
Asia	KR	42	42	42
Control	CA	21	20	19
Control	AU	42	42	19
Control	NZ	41	41	18
Control	IL	42	42	42
Systemic	US + GB	22	20	19
Systemic	JP + HK + KR	21	19	18
Systemic	IT + ES + IE	20	18	17

Note: Buffer allocations are heterogeneous based on node centrality.
Core nodes: 0.03/0.05 (Layer/Total); Periphery nodes: 0.08/0.12. Network size $N = 43$.

Table 2: MultiRank Centrality Scores and Rankings of All 43 Global Nodes (2007–2009)

Country / Node	**2007**		**2008**		**2009**	
	Rank	**Score**	**Rank**	**Score**	**Rank**	**Score**
United Kingdom	1	0.1976	1	0.1832	1	0.1819
United States	2	0.1295	2	0.1284	2	0.1277
France	3	0.0686	4	0.0647	4	0.0636
Germany	4	0.0575	3	0.0650	3	0.0667
Netherlands	5	0.0455	5	0.0460	5	0.0438
Italy	6	0.0412	8	0.0405	8	0.0411
Ireland	7	0.0370	6	0.0445	7	0.0423
Switzerland	8	0.0369	10	0.0295	11	0.0258
Spain	9	0.0349	7	0.0432	6	0.0430
Japan	10	0.0295	11	0.0279	10	0.0312
Belgium	11	0.0279	12	0.0218	12	0.0214
Luxembourg	12	0.0277	9	0.0354	9	0.0325
Canada	13	0.0198	13	0.0202	13	0.0211
Norway	14	0.0196	14	0.0193	15	0.0184
Sweden	15	0.0181	15	0.0188	14	0.0191
Denmark	16	0.0155	16	0.0161	16	0.0184
Australia	17	0.0118	17	0.0125	17	0.0142
Portugal	18	0.0116	20	0.0112	19	0.0120
Austria	19	0.0113	21	0.0107	21	0.0102
China	20	0.0108	25	0.0083	24	0.0092
Finland	21	0.0097	19	0.0113	18	0.0129
Russia	22	0.0096	18	0.0114	22	0.0099
Greece	23	0.0092	22	0.0099	20	0.0105
Brazil	24	0.0091	23	0.0090	23	0.0093
Korea	25	0.0091	27	0.0078	26	0.0082
Türkiye	26	0.0074	24	0.0083	27	0.0081
Mexico	27	0.0069	29	0.0073	31	0.0070
India	28	0.0067	28	0.0073	28	0.0076
New Zealand	29	0.0067	31	0.0061	29	0.0074
Poland	30	0.0067	26	0.0082	25	0.0085
Iceland	31	0.0059	35	0.0053	35	0.0053
Hungary	32	0.0059	30	0.0072	30	0.0072
Saudi Arabia	33	0.0057	32	0.0060	32	0.0065
South Africa	34	0.0055	34	0.0056	34	0.0055
Israel	35	0.0054	38	0.0047	38	0.0049
Chile	36	0.0054	33	0.0057	33	0.0058
Argentina	37	0.0052	36	0.0051	37	0.0049
Indonesia	38	0.0052	37	0.0051	36	0.0051
Colombia	39	0.0049	39	0.0047	39	0.0046
Costa Rica	40	0.0045	40	0.0044	40	0.0044
Hong Kong SAR	41	0.0044	41	0.0043	41	0.0043
Chinese Taipei	42	0.0044	42	0.0043	42	0.0043
Macao SAR	43	0.0044	43	0.0043	43	0.0043

Table 5: Impact of Credit Crunch on Network Survivability (2008 Q4)

Category	**Victim (Shock Origin)**	**Final Surviving Nodes** ($N = 43$)		
		Crunch = 0	**Crunch = 0.3**	**Crunch = 0.5**
Global	US	20	3	3
Global	GB	20	1	1
EU Core	DE	20	1	1
EU Core	FR	20	1	1
EU	IT	19	19	19
EU	ES	19	19	19
Hub	CH	20	1	1
Hub	LU	20	1	1
Hub	IE	20	2	2
Asia	JP	20	1	1
Asia	HK	42	42	42
Asia	KR	42	1	1
Control	CA	20	16	16
Control	AU	42	1	1
Control	NZ	41	41	41
Control	IL	42	42	42
Systemic	US + GB	20	1	1
Systemic	JP + HK + KR	19	0*	0*
Systemic	IT + ES + IE	18	2	2

Note: Simulations conducted on the 2008 Q4 network topology.

* Denotes a complete systemic wipeout (0 survivors).

Table 6: Detailed Intervention Outcomes: Topological Survival vs. Economic Mass Preserved

Category	Shock Origin	Bailout Target	Survivors	Net Saved	Asset Preserved (%)
	US	None (Baseline)	3	-	-
	US	US (Direct Save)	43	40	100.00%
GFC Core	US	GB (Firewall attempt)	4	1	27.50%
	US	DE	4	1	9.70%
	US	JP	4	1	6.50%
	GB	None (Baseline)	1	-	-
European Core	GB	GB (Direct Save)	43	42	100.00%
	GB	US (Firewall attempt)	2	1	12.00%
	DE	None (Baseline)	1	-	-
	DE	DE (Direct Save)	43	42	100.00%
	DE	GB	2	1	27.50%
	US+GB	None (Baseline)	1	-	-
Transatlantic	US+GB	GB	4	3	27.50%
	US+GB	US	2	1	12.00%
	JP	None (Baseline)	1	-	-
Japan Shock	JP	JP (Direct Save)	43	42	100.00%
	JP	US	2	1	12.00%
	JP	HK+KR (Asian Shield)	2	1	0.40%
	JP+HK+KR	None (Baseline)	0	-	-
Asian Crisis	JP+HK+KR	JP+HK+KR (Direct Save)	43	43	100.00%
	JP+HK+KR	GB (Save the West)	1	1	27.50%
	JP+HK+KR	US (Save the West)	1	1	12.00%
	IT+ES+IE	None (Baseline)	2	-	-
Euro Debt	IT+ES+IE	IT+ES+IE (Direct Save)	43	41	100.00%
	IT+ES+IE	DE+FR (Core Shield)	4	2	21.20%

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Table 6 – continued from previous page

Category	**Shock Origin**	**Bailout Target**	**Survivors**	**Net Saved**	**Asset Preserved (%)**
	IT+ES+IE	IT+ES (Partial Save)	4	2	0.00%
	AU	None (Baseline)	1	-	-
	AU	AU (Direct Save)	43	42	100.00%
Aussie Shock	AU	GB	2	1	27.50%
	AU	US	2	1	12.00%
	AU	IT+ES (Euro Shield)	3	2	0.00%

Note: Simulations performed on 2008 Q4 multiplex network ($N = 43$). "Net Saved" represents nodes preserved relative to the Baseline. "Asset Preserved" indicates the global economic mass protected from systemic collapse.

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